TABLE Summary of heat transfer processes

Mode	Mechanism(s)	Rate Equation
Conduction	Diffusion of energy due to random molecular motion	$q''_x\ (\mathrm{W/m^2}) = -k\dfrac{dT}{dx}$
Convection	Diffusion of energy due to random molecular motion plus energy transfer due to bulk motion (advection)	$q''\ (\mathrm{W/m^2}) = h(T_s - T_\infty)$
Radiation	Energy transfer by electromagnetic waves	$q''(\mathrm{W/m^2}) = \varepsilon\sigma(T_s^4 - T_{sur}^4)$ or $q\ (\mathrm{W}) = h_r A(T_s - T_{sur})$

TABLE One-dimensional, steady-state solutions to the heat equation with no generation

	Plane Wall	Cylindrical Wall	Spherical Wall
Heat equation	$\dfrac{d^2T}{dx^2} = 0$	$\dfrac{1}{r}\dfrac{d}{dr}\left(r\dfrac{dT}{dr}\right) = 0$	$\dfrac{1}{r^2}\dfrac{d}{dr}\left(r^2\dfrac{dT}{dr}\right) = 0$
Temperature distribution	$T_{s,1} - \Delta T\dfrac{x}{L}$	$T_{s,2} + \Delta T\dfrac{\ln(r/r_2)}{\ln(r_1/r_2)}$	$T_{s,1} - \Delta T\left[\dfrac{1-(r_1/r)}{1-(r_1/r_2)}\right]$
Heat flux (q'')	$k\dfrac{\Delta T}{L}$	$\dfrac{k\,\Delta T}{r\ln(r_2/r_1)}$	$\dfrac{k\,\Delta T}{r^2[(1/r_1)-(1/r_2)]}$
Heat rate (q)	$kA\dfrac{\Delta T}{L}$	$\dfrac{2\pi Lk\,\Delta T}{\ln(r_2/r_1)}$	$\dfrac{4\pi k\,\Delta T}{(1/r_1)-(1/r_2)}$
Thermal resistance ($R_{t,cond}$)	$\dfrac{L}{kA}$	$\dfrac{\ln(r_2/r_1)}{2\pi Lk}$	$\dfrac{(1/r_1)-(1/r_2)}{4\pi k}$

TABLE Thermophysical Properties of Gases at Atmospheric Pressure

T (K)	ρ (kg/m^3)	c_p (kJ/kg · K)	$\mu \cdot 10^7$ (N · s/m^2)	$\nu \cdot 10^6$ (m^2/s)	$k \cdot 10^3$ (W/m · K)	$\alpha \cdot 10^6$ (m^2/s)	Pr
Air							
100	3.5562	1.032	71.1	2.00	9.34	2.54	0.786
150	2.3364	1.012	103.4	4.426	13.8	5.84	0.758
200	1.7458	1.007	132.5	7.590	18.1	10.3	0.737
250	1.3947	1.006	159.6	11.44	22.3	15.9	0.720
300	1.1614	1.007	184.6	15.89	26.3	22.5	0.707
350	0.9950	1.009	208.2	20.92	30.0	29.9	0.700
400	0.8711	1.014	230.1	26.41	33.8	38.3	0.690
450	0.7740	1.021	250.7	32.39	37.3	47.2	0.686
500	0.6964	1.030	270.1	38.79	40.7	56.7	0.684
550	0.6329	1.040	288.4	45.57	43.9	66.7	0.683
600	0.5804	1.051	305.8	52.69	46.9	76.9	0.685
650	0.5356	1.063	322.5	60.21	49.7	87.3	0.690
700	0.4975	1.075	338.8	68.10	52.4	98.0	0.695
750	0.4643	1.087	354.6	76.37	54.9	109	0.702
800	0.4354	1.099	369.8	84.93	57.3	120	0.709

TABLE Selected Dimensionless Groups of Heat Transfer

Group	Definition	Interpretation
Biot number (Bi)	$\frac{hL}{k_s}$	Ratio of the internal thermal resistance of a solid to the boundary layer thermal resistance.
Bond number (Bo)	$\frac{g(\rho_l - \rho_v)L^2}{\sigma}$	Ratio of gravitational and surface tension forces.
Coefficient of friction (C_f)	$\frac{\tau_s}{\rho V^2/2}$	Dimensionless surface shear stress.
Eckert number (Ec)	$\frac{V^2}{c_p(T_s - T_\infty)}$	Kinetic energy of the flow relative to the boundary layer enthalpy difference.
Fourier number (Fo)	$\frac{\alpha t}{L^2}$	Ratio of the heat conduction rate to the rate of thermal energy storage in a solid. Dimensionless time.
Friction factor (f)	$\frac{\Delta p}{(L/D)(\rho u_m^2/2)}$	Dimensionless pressure drop for internal flow.
Grashof number (Gr_L)	$\frac{g\beta(T_s - T_\infty)L^3}{\nu^2}$	Ratio of buoyancy to viscous forces.
Colburn j factor (j_H)	$St\ Pr^{2/3}$	Dimensionless heat transfer coefficient.
Jakob number (Ja)	$\frac{c_p(T_s - T_{sat})}{h_{fg}}$	Ratio of sensible to latent energy absorbed during liquid–vapor phase change.
Nusselt number (Nu_L)	$\frac{hL}{k_f}$	Dimensionless temperature gradient at the surface.
Peclet number (Pe_L)	$\frac{VL}{\alpha} = Re_L Pr$	Dimensionless independent heat transfer parameter.
Prandtl number (Pr)	$\frac{c_p\mu}{k} = \frac{\nu}{\alpha}$	Ratio of the momentum and thermal diffusivities.
Reynolds number (Re_L)	$\frac{VL}{\nu}$	Ratio of the inertia and viscous forces.
Stanton number (St)	$\frac{h}{\rho V c_p} = \frac{Nu_L}{Re_L Pr}$	Modified Nusselt number.
Weber number (We)	$\frac{\rho V^2 L}{\sigma}$	Ratio of inertia to surface tension forces.

$$Ra_L = Gr_L \Pr = \frac{g\beta (T_s - T_\infty)x^3}{\nu\alpha}, \qquad \beta = -\frac{1}{T}$$

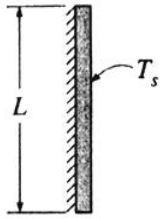
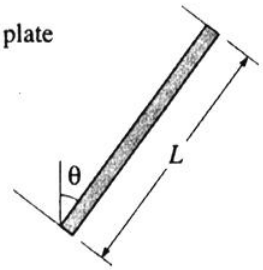
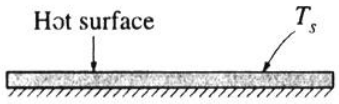
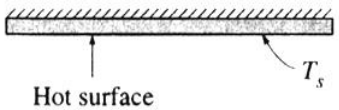
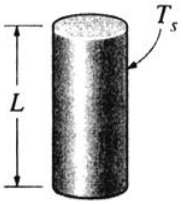
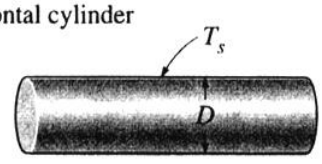
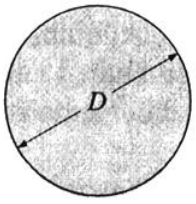
TABLE Summary of convection heat transfer correlations for external flow

Correlation	Geometry	Conditions
$\delta = 5x\,Re_x^{-1/2}$	Flat plate	Laminar, T_f
$C_{f,x} = 0.664Re_x^{-1/2}$	Flat plate	Laminar, local, T_f
$Nu_x = 0.332Re_x^{1/2}\,Pr^{1/3}$	Flat plate	Laminar, local, T_f, $0.6 \lesssim Pr \lesssim 50$
$\delta_t = \delta\,Pr^{-1/3}$	Flat plate	Laminar, T_f
$\overline{C}_{f,x} = 1.328Re_x^{-1/2}$	Flat plate	Laminar, average, T_f
$\overline{Nu}_x = 0.664Re_x^{1/2}\,Pr^{1/3}$	Flat plate	Laminar, average, T_f, $0.6 \lesssim Pr \lesssim 50$
$Nu_x = 0.565Pe_x^{1/2}$	Flat plate	Laminar, local, T_f, $Pr \lesssim 0.05$
$C_{f,x} = 0.0592Re_x^{-1/5}$	Flat plate	Turbulent, local, T_f, $Re_x \lesssim 10^8$
$\delta = 0.37x\,Re_x^{-1/5}$	Flat plate	Turbulent, local, T_f, $Re_x \lesssim 10^8$
$Nu_x = 0.0296Re_x^{4/5}\,Pr^{1/3}$	Flat plate	Turbulent, local, T_f, $Re_x \lesssim 10^8$, $0.6 \lesssim Pr \lesssim 60$
$\overline{C}_{f,L} = 0.074Re_L^{-1/5} - 1742Re_L^{-1}$	Flat plate	Mixed, average, T_f, $Re_{x,c} = 5 \times 10^5$, $Re_L \lesssim 10^8$
$\overline{Nu}_L = (0.037Re_L^{4/5} - 871)Pr^{1/3}$	Flat plate	Mixed, average, T_f, $Re_{x,c} = 5 \times 10^5$, $Re_L \lesssim 10^8$, $0.6 < Pr < 60$
$\overline{Nu}_D = C\,Re_D^m\,Pr^{1/3}$	Cylinder	Average, T_f, $0.4 < Re_D < 4 \times 10^5$, $Pr \gtrsim 0.7$
$\overline{Nu}_D = C\,Re_D^m\,Pr^n(Pr/Pr_s)^{1/4}$	Cylinder	Average, T_∞, $1 < Re_D < 10^6$, $0.7 < Pr < 500$
$\overline{Nu}_D = 0.3 + \{0.62Re_D^{1/2}\,Pr^{1/3} \times [1 + (0.4/Pr)^{2/3}]^{-1/4}\} \times [1 + (Re_D/282{,}000)^{5/8}]^{4/5}$	Cylinder	Average, T_f, $Re_D\,Pr > 0.2$
$\overline{Nu}_D = 2 + (0.4Re_D^{1/2} + 0.06Re_D^{2/3})Pr^{0.4} \times (\mu/\mu_s)^{1/4}$	Sphere	Average, T_∞, $3.5 < Re_D < 7.6 \times 10^4$, $0.71 < Pr < 380$
$\overline{Nu}_D = 1.13C_1Re_{D,\max}^m\,Pr^{1/3}$	Tube bank[b]	Average, $\overline{T}_f$, $2000 < Re_{D,\max} < 4 \times 10^4$, $Pr \gtrsim 0.7$
$\overline{Nu}_D = C\,Re_{D,\max}^m\,Pr^{0.36}(Pr/Pr_s)^{1/4}$	Tube bank[b]	Average, $\overline{T}$, $1000 < Re_{D,\max} < 2 \times 10^6$, $0.7 < Pr < 500$

TABLE Summary of convection correlations for flow in a circular tube

Correlation	Conditions
$f = 64/Re_D$	Laminar, fully developed
$Nu_D = 4.36$	Laminar, fully developed, uniform q_s'', $Pr \gtrsim 0.6$
$Nu_D = 3.66$	Laminar, fully developed, uniform T_s, $Pr \gtrsim 0.6$
$\overline{Nu}_D = 3.66 + \dfrac{0.0668(D/L)Re_D Pr}{1 + 0.04[(D/L)Re_D Pr]^{2/3}}$	Laminar, thermal entry length ($Pr \gg 1$ or an unheated starting length), uniform T_s
or	
$\overline{Nu}_D = 1.86\left(\dfrac{Re_D Pr}{L/D}\right)^{1/3}\left(\dfrac{\mu}{\mu_s}\right)^{0.14}$	Laminar, combined entry length $\{[Re_D Pr/(L/D)]^{1/3}(\mu/\mu_s)^{0.14}\} \gtrsim 2$, uniform T_s, $0.48 < Pr < 16{,}700$, $0.0044 < (\mu/\mu_s) < 9.75$
$f = 0.316Re_D^{-1/4}$	Turbulent, fully developed, $Re_D \lesssim 2 \times 10^4$
$f = 0.184Re_D^{-1/5}$	Turbulent, fully developed, $Re_D \gtrsim 2 \times 10^4$
or	
$f = (0.790 \ln Re_D - 1.64)^{-2}$	Turbulent, fully developed, $3000 \lesssim Re_D \lesssim 5 \times 10^6$
$Nu_D = 0.023Re_D^{4/5} Pr^n$	Turbulent, fully developed, $0.6 \leq Pr \leq 160$, $Re_D \geq 10{,}000$, $(L/D) \gtrsim 10$, $n = 0.4$ for $T_s > T_m$ and $n = 0.3$ for $T_s < T_m$
or	
$Nu_D = 0.027Re_D^{4/5} Pr^{1/3}\left(\dfrac{\mu}{\mu_s}\right)^{0.14}$	Turbulent, fully developed, $0.7 \leq Pr \leq 16{,}700$, $Re_D \gtrsim 10{,}000$, $L/D \gtrsim 10$
or	
$Nu_D = \dfrac{(f/8)(Re_D - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}$	Turbulent, fully developed, $0.5 < Pr < 2000$, $3000 \lesssim Re_D \lesssim 5 \times 10^6$, $(L/D) \gtrsim 10$
$Nu_D = 4.82 + 0.0185(Re_D Pr)^{0.827}$	Liquid metals, turbulent, fully developed, uniform q_s'', $3.6 \times 10^3 < Re_D < 9.05 \times 10^5$, $10^2 < Pe_D < 10^4$
$Nu_D = 5.0 + 0.025(Re_D Pr)^{0.8}$	Liquid metals, turbulent, fully developed, uniform T_s, $Pe_D > 100$

Empirical correlations for the average Nusselt number for natural convection over surfaces

Geometry	Characteristic length L_c	Range of Ra	Nu
Vertical plate 	L	10^4–10^9 10^9–10^{13} Entire range	$Nu = 0.59Ra_L^{1/4}$ $Nu = 0.1Ra_L^{1/3}$ $Nu = \left\{0.825 + \frac{0.387Ra_L^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}}\right\}^2$ (complex but more accurate)
Inclined plate 	L		Use vertical plate equations for the upper surface of a cold plate and the lower surface of a hot plate Replace g by $g\cos\theta$ for $Ra < 10^9$
Horiontal plate (Surface area A and perimeter p) (a) Upper surface of a hot plate (or lower surface of a cold plate)  (b) Lower surface of a hot plate (or upper surface of a cold plate) 	A_s/p	10^4–10^7 10^7–10^{11} 10^5–10^{11}	$Nu = 0.54Ra_L^{1/4}$ $Nu = 0.15Ra_L^{1/3}$ $Nu = 0.27Ra_L^{1/4}$
Vertical cylinder 	L		A vertical cylinder can be treated as a vertical plate when $D \geq \frac{35L}{Gr_L^{1/4}}$
Horizontal cylinder 	D	$Ra_D \leq 10^{12}$	$Nu = \left\{0.6 + \frac{0.387Ra_D^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}}\right\}^2$
Sphere 	D	$Ra_D \leq 10^{11}$ ($Pr \geq 0.7$)	$Nu = 2 + \frac{0.589Ra_D^{1/4}}{[1 + (0.469/Pr)^{9/16}]^{4/9}}$

$$Ra_L = Gr_L \Pr = \frac{g\beta (T_s - T_\infty)x^3}{\nu\alpha}$$